Deep Transfer Learning for Kidney Disease Detection Using CT Scan Images

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***Abstract*— Kidney disease is a significant health issue that leads to a high number of deaths worldwide. Accurate and timely detection of kidney diseases, including cysts, stones, and tumours, is critical for effective treatment and patient outcomes. Deep learning methodologies, specifically transfer learning, have recently been widely used in medical image analysis. This study comprehensively evaluates seven transfer learning models—Xception, DenseNet201, MobileNet, InceptionV3, VGG16, ResNet50, and EfficientNetB0—for multiclass kidney disease classification using CT scan images. The models are assessed based on key performance metrics such as accuracy, loss, precision, recall, F1-score, and AUC. Results indicate that ResNet50 delivers the best overall performance, making it a promising model for clinical applications. By comparing these state-of-the-art models, this research provides valuable insights into the most effective transfer learning approaches for kidney disease detection, paving the way for robust computer-aided diagnostic systems.**

Keywords—Kidney Disease, Deep Learning, Transfer Learning, CT Scan Images.

I. Introduction

Kidney diseases, including conditions such as cysts, stones, and tumours, present significant challenges to healthcare systems worldwide. Accurate and timely diagnosis of these conditions is crucial for effective treatment and improving patient outcomes. Medical imaging, particularly computed tomography (CT) scans, plays a vital role in diagnosing kidney diseases by providing detailed information about the structural abnormalities in the kidneys [1]. However, the manual interpretation of CT images is often time-consuming, subject to variability among radiologists, and prone to diagnostic errors. These challenges highlight the need for automated, computer-aided diagnostic (CAD) systems to assist clinicians in detecting and classifying kidney diseases more accurately and efficiently [2].

Deep learning, specifically convolutional neural networks (CNNs), has substantially advanced medical image analysis in recent years. CNNs have demonstrated exceptional performance in tasks such as image classification, segmentation, and object detection. In kidney disease identification and prediction also, deep learning has been widely used [3]. Despite these advancements, training deep learning models from scratch requires large labelled datasets and considerable computational power, which are often scarce in medical imaging contexts. To address these limitations, transfer learning has emerged as an effective approach [4]. By leveraging pre-trained models on large datasets, transfer learning enables fine-tuning these models for specific tasks, such as kidney disease classification, with smaller medical datasets and reduced training times [5]. This technique has been widely adopted for various medical imaging tasks, significantly improving model performance and reducing training time [6] [7].

This study aims to conduct a comprehensive performance analysis of seven state-of-the-art transfer learning models — Xception [8], DenseNet201 [9], VGG16 [10], ResNet50 [11], EfficientNetB0 [12], MobileNet [13] and InceptionV3 [14] — in classifying kidney diseases, including cysts, stones, and tumours, using CT scan images. These models have been chosen for their strong track record in image classification tasks and ability to extract high-level features critical for accurate medical diagnoses. However, there has been limited research comparing their performance specifically for multiclass kidney disease prediction.

To address this gap, we conduct a comprehensive analysis of these transfer learning models on a multiclass kidney disease dataset. Each model is evaluated using performance metrics such as accuracy, precision, recall, F1-score, and area under the curve (AUC). This study aims to determine which model performs best in terms of classification accuracy and practical applicability in real-world clinical settings. By identifying the most effective model, this research aims to contribute to developing reliable, automated diagnostic systems for kidney disease detection, supporting healthcare professionals in their decision-making processes.

In the following sections, we report a literature review and describe the dataset, methodology, and experimental setup used for evaluating the transfer learning models, followed by a detailed analysis of the results and a discussion of the findings.

II. Literature Review

The application of computer-aided diagnosis systems for identifying and predicting kidney disease through medical imaging has shown increasing effectiveness in recent years. Deep learning methods, particularly transfer learning techniques, have gained prominence for their reliability, consistency, and precision in terms of outcomes [15] [16]. Several studies have explored using CNNs and transfer learning for kidney disease classification.

Ma et al. [17] proposed a heterogeneous modified artificial neural network (HMANN) for the early detection and diagnosis of chronic renal failure. Chen et al. [18] developed an adaptive hybridized deep CNN (HDCNN) for efficient kidney disease detection. Knight et al. [19] trained a U-Net model for kidney segmentation in CT scans and achieved a Sørensen–Dice coefficient of 0.9620. Kwon et al. [20] enhanced machine learning accuracy for disease diagnosis using various MRI (magnetic resonance imaging) sequences and achieved high accuracy with Dice coefficients of 0.951 and 0.952 for different sequences. Elbedwehy et al. [21]

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developed an advanced AI-driven diagnostic system that combined AlexNet and ConvNeXt architectures, achieving an exceptional classification accuracy of 99.85%. Kumar et al. [22] presented a novel deep-learning model combining a fuzzy deep neural network (DNN), which demonstrated an accuracy of 99.23% in recognizing and predicting kidney disease. Hossain et al. [23] employed three CNN classification methods based on watershed segmentation and DNNs to classify four types of kidney CT images: cyst, normal, stone, and tumour. Their proposed CNN model achieved the highest accuracy of 98.66% on the test set, outperforming the EANet and ResNet50. Bingol et al. [24] introduced a hybrid model for classifying kidney CT images containing stones, tumours, and cysts. Their proposed 29-layer model combined feature maps from specific convolution layers, optimized them using the relief method, and employed a wide neural network classifier. This approach achieved an impressive accuracy of 99.37%, surpassing that of eight pre-trained models. Islam et al. [25] focused on three major kidney disease categories: kidney stones, cysts, and tumours. They evaluated six machine learning models, including state-of-the-art vision transformers and well-known deep learning models. The swine transformer outperformed the other models with an accuracy of 99.30% and demonstrated a superior training speed.

Qadir and Abd [26] employed a hybrid approach that combined pre-trained models for feature extraction and machine-learning algorithms for classification. The Densenet201 - random forest approach achieved an impressive accuracy of 99.719% for kidney disease image diagnosis. Bhattacharjee et al. [1] proposed a modified Xception DNN-based model that attained 100% accuracy, F1 score, recall, and precision for kidney disease multiclass classification. This model outperformed the original Xception model and the existing methods. Canbay et al. [27] presented a privacy-preserving transfer-learning framework for kidney disease classification. They utilized various CNN architectures including InceptionResNetV2, DenseNet201, InceptionV3, Xception, ResNet50, MobileNet, and VGG19. The MobileNet architecture stood out with an exceptional accuracy of 99.83% for privacy-preserving classification. Rana et al. [28] introduced the "KidneyMultiNet" framework, combining DenseNet201 and Xception models. This approach achieved a maximum performance of 99.92% in predicting kidney disease based on the CT scan datasets. Majid et al. [29] investigated the use of machine learning and fine-tuned transfer learning for kidney tumour detection. They compared various detection approaches and found that the fine-tuned ResNet101 and DenseNet121 models achieved the highest accuracy of 94.09% when using 32 × 32 input images. Saxena et al. [30] compared seven transfer learning methods for classifying four classes of kidney disease. Among these, EfficientNetB0 demonstrated the highest accuracy of 96.02%.

Despite significant advancements in applying transfer learning models for kidney disease detection, several research gaps remain. While numerous studies have demonstrated high accuracy in classifying kidney conditions using CT scan images, the majority focus on specific models or architectures without providing comprehensive comparisons across a broader range of transfer learning models. Moreover, studies often emphasize accuracy but neglect to explore other vital metrics such as computational efficiency, scalability, and model interpretability in real-world clinical settings.

The current study aims to fill these gaps by conducting an extensive performance analysis of seven leading transfer learning models of different characteristics on a multiclass kidney disease dataset. In addition to accuracy, this research will evaluate the models based on several other performance metrics, providing a holistic view of their potential for practical deployment in kidney disease detection using CT scan images.

III. Methodology

For the experiment, a publicly accessible dataset [1] containing 12,446 images of four classes of patients diagnosed with having a kidney cyst, stone, tumour, or normal findings [25], as shown in TABLE I. Sample images of each class are shown in Fig. 1.

TABLE I. Detailed sample-wise distribution of the CT scan images datasets

Dataset	Parameters	No. of images	Class	Label
Original	Cyst	3709	0	1
	Stone	1377	2	2
	Tumour	2283	3	3
	Normal	5077	1	4
Up/down-sampled	Cyst	2505	0	1
	Stone	1977	2	2
	Tumour	2283	3	3
	Normal	2403	1	4

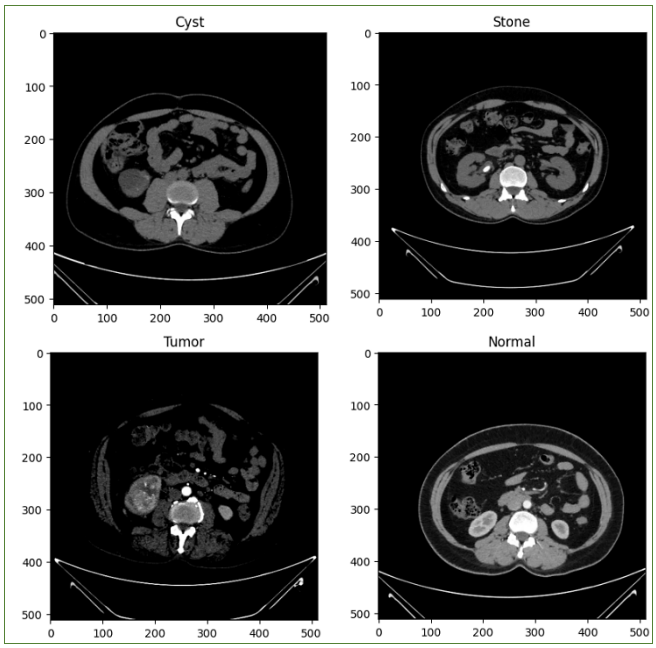

Fig. 1. Sample images of each class of kidney dataset

For efficient training of the considered model, the dataset was class-wise up or down-sampled, resulting in a total of 9168 images. The details of data augmentation parameters and

[1] https://www.kaggle.com/datasets/nazmul0087/ct-kidney-dataset-normal-cyst-tumor-and-stone/data

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their values are shown in TABLE II. Furthermore, the dataset was split into three portions: training (70%), validation (10%), and testing (20%).

TABLE II. DATA AUGMENTATION

Particular	Value
Shear range	0.2
Zoom range	0.2
Rotation range	24
Vertical flip	TRUE
Horizontal flip	TRUE
Vertical shearing	0.2
Horizontal shearing	0.2

TABLE III. SUMMARY OF HYPERPARAMETER VALUES

Hyperparameter	Value
Optimizer 1	Adam
Optimizer 2	RMSprop
Learning rate	0.001
Loss	Categorical cross-entropy
Dropout probability	0.4
Batch size	32
Number of epochs	100

For the comparative analysis purpose, we chose seven transfer learning models of different characteristics, as shown in Fig. 2. All the models were trained with the augmented training data and subsequently validated and tested. TABLE III shows the suitable hyperparameters and their values used to train the models.

IV. PERFORMANCES EVALUATION

After training the models with 6417 samples and validating with 908 samples, we tested each model on 1843 samples. The correct and incorrect classifications of each model are shown in Fig. 3 and Fig. 4, respectively. Among all the models, RestNet50 performed best, while Xception performed worst in both cases. For example, out of 1843 instances, RestNet50 correctly classified 1809 times and failed 34 times, whereas Xception correctly classified merely 1312 times and failed 531 times. Six metrics were used to further evaluate the models' performance.

The average test accuracies and loss of each model for each class are shown in Fig. 5 and Fig. 6. The score for accuracy represents the proportion of correct detections, and values representing a loss indicate the model's learning progress and error minimization. It is confirmed that ResNet50 had the best accuracy and loss, with 98.12% and 0.07, respectively. Xception had worst accuracy with 70.18%, and VGG16 had the worst loss with 0.84, respectively, were the best and worst performers, respectively.

The precision and recall of each model are shown in Fig. 7 and Fig. 8, respectively. Precision and recall provide insights into the trade-off between correctly identifying disease and avoiding false alarms. ResNet50 achieved the highest average precision (98%) and recall (97.75%), whereas Xception was again the lowest scorer in both metrics, with 74% and 72%, respectively. Fig. 9 presents the models' F1-scores, which offer a combined view of precision and recall. Expectedly, ResNet50 attained the highest average F1-score of 97.75%, and Xception got a mere 71.75%.

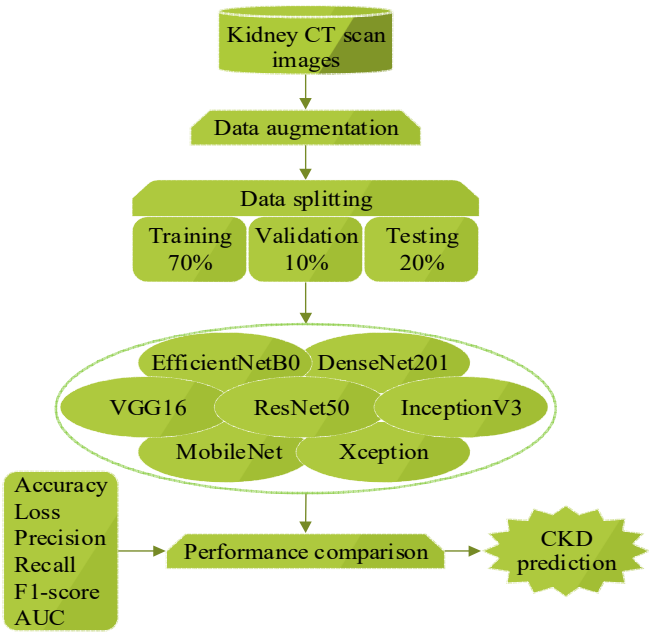

Fig. 2. Workflow of this performance evaluation and analysis study

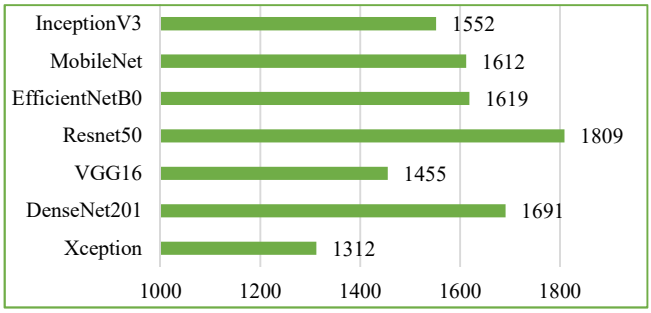

Fig. 3. Correct classifications

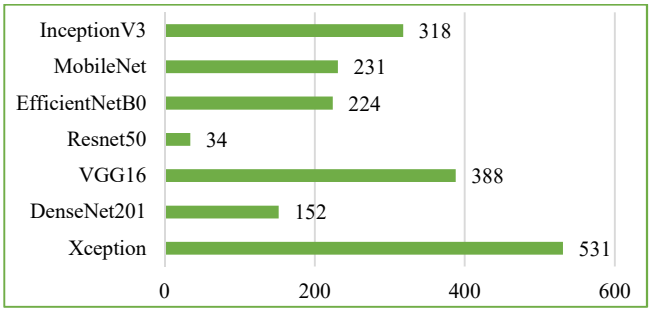

Fig. 4. Incorrect classifications

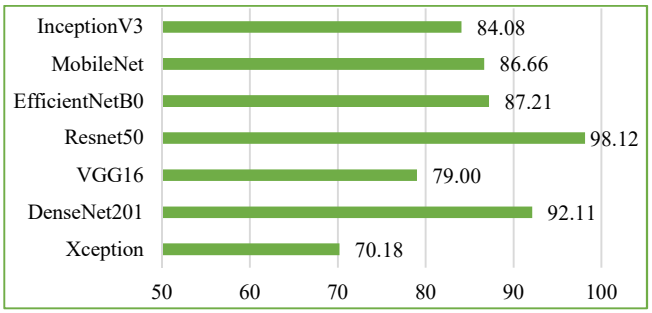

Fig. 5. Accuracy comparison of the models

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The AUC measures shown in Fig. 10 indicate that ResNet50 is more capable of discriminating between classes with an AUC value of 99.33%. In contrast, Xception exhibited the least class-discriminating capability with an AUC value of 87.43%.

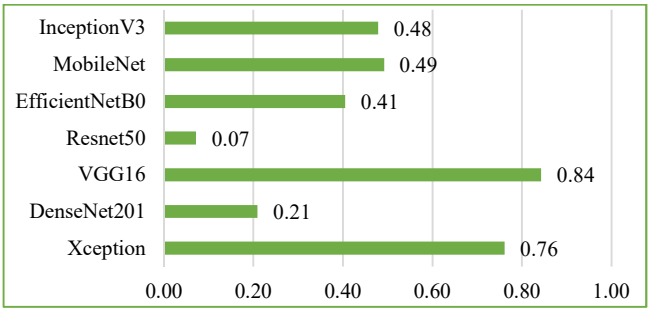

Fig. 6. Loss comparison of the models

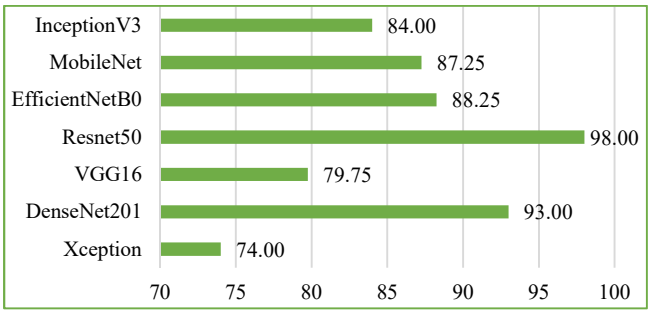

Fig. 7. Precision comparison of the models

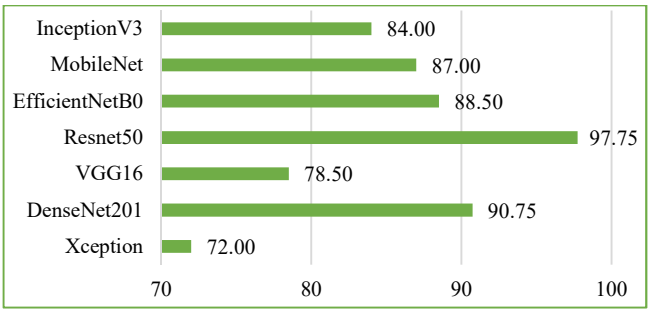

Fig. 8. Recall comparison of the models

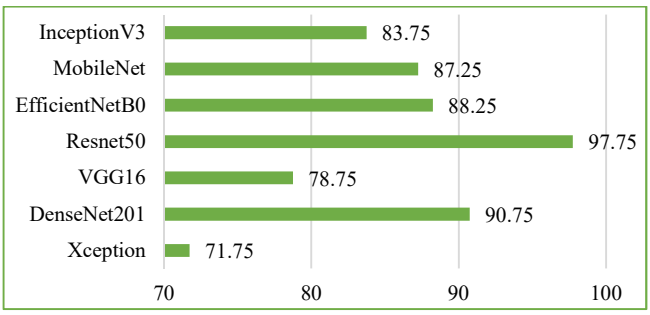

Fig. 9. F1-score comparison of the models

TABLE IV ranks the best and worst-performing models against each metric to summarise the performance comparison. ResNet50 emerged as the best performer in each metric, followed by DenseNet201 and EfficientNetB0. On the other hand, Xception has been the worst performer in each metric except for the loss. VGG16 has been the second-worst performer after Xception. Because ResNet50 did the best overall at identifying kidney disease in our study, to see how it stacked up in the literature, we compared it with a few similar papers, as shown in Table V.

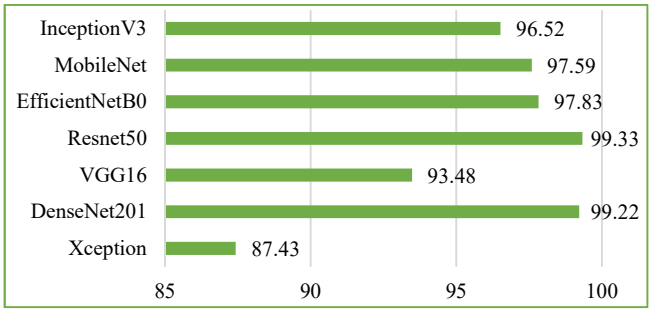

Fig. 10. AUC comparison of the models

TABLE IV. RANKING OF THE CONSIDERED TRANSFER LEARNING MODELS AS PER THEIR AVERAGE PERFORMANCES (1: BEST, 7: WORST)

Model	Accu-racy	Loss	Preci-sion	Recall	F1-score	AUC
Xception	7	6	7	7	7	7
DenseNet201	2	2	2	2	2	2
VGG16	6	7	6	6	6	6
Resnet50	1	1	1	1	1	1
EfficientNetB0	3	3	3	3	3	3
MobileNet	4	5	4	4	4	4
InceptionV3	5	4	5	5	5	5

V. CONCLUSION

This study conducted a comparative performance analysis using CT scan images for kidney disease classification of seven transfer learning models — Xception, DenseNet201, MobileNet, InceptionV3, EfficientNetB0, ResNet50, and VGG16. The evaluation was based on a comprehensive set of performance metrics, including accuracy, precision, recall, F1-score, AUC, and loss. Among the models, ResNet50 consistently demonstrated superior performance across most metrics, achieving the highest accuracy (98.12%), precision (98%), recall (97.75%), and F1-score (97.75%), indicating its strong potential as a reliable tool for kidney disease detection in clinical settings. While DenseNet201 also performed well, Xception and VGG16 underperformed, with Xception achieving the lowest accuracy (70.18%) and VGG16 showing the highest loss (0.84).

Timely and accurate detection of kidney disease can significantly impact patient outcomes. This comparison analysis provides valuable insights into the strengths and limitations of each model, helping healthcare practitioners and researchers choose the best transfer learning model for kidney disease identification. The findings from this study can aid in developing robust computer-aided diagnostic systems, supporting healthcare professionals in their decision-making processes and ultimately leading to improved patient care. The superior performance of ResNet50 positions it as a promising candidate for real-world clinical applications, contributing to more efficient and accurate diagnosis.

While this study focused on transfer learning models, further research could explore alternative architectures or ensemble methods to enhance performance. A systematical comparison using the benchmark datatsets will provide more information and insights. Additionally, applying explainable AI techniques would improve the interpretability of the

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models, enabling better insights into their decision-making processes. Future work could also extend this analysis to multiple datasets to ensure the models' generalizability across different populations, disease subgroups, and imaging sources.

TABLE V. COMPARATIVE ANALYSIS OF THE STUDIED MODEL WITH EXISTING SIMILAR WORKS

Reference	Models	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC (%)
Ma et al. [17]	HMANN, SVM, MLP with backpropagation	CT kidney dataset	97.50 with integration of all	-	-	-	-
Chen at al. [18]	AHDCNN	NIH's DeepLesion dataset	97.14	93	77	84.2	1.0
Kwon et al. [20]	U-Net	MRI (FIESTA and SSFSE)	95.2	93.67	94	93.62	-
Majid et al. [29]	RF, SVM, GB, LGBM, ResNet101, DenseNet121	CT kidney dataset	98.22	95.77	93.23	98.43	97.73
Saxena et al. [30]	EfficientNetB0, Xception, VGG16, ResNet50, MobileNet, InceptionV3, DenseNet121	Collected dataset	96.02	94.22	95.13	94.70	-
This paper	Xception, DenseNet201, MobileNet, VGG16, InceptionV3, ResNet50, EfficientNetB0	CT kidney dataset	98.12 with Resnet50	98	97.75	97.75	99.33

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